



Short communication

Modeling growth rates as a function of temperature: Model performance evaluation with focus on the suboptimal temperature range

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ABSTRACT

Secondary models, describing the microbial growth rate as a function of temperature, are evaluated with focus on model performance in the suboptimal temperature region. *Escherichia coli* K12 MG1655 is considered as the case study. A large set of $\sqrt{\mu_{\max}(T)}$ -estimates is fitted with (1) the cardinal temperature model with inflection (CTMI, Rosso et al., 1993), (2) the square root model (SQRT, Ratkowsky et al., 1983), and (3) the CTMI adapted to describe the particular behavior of *Listeria* at suboptimal temperatures (aCTMI, Le Marc et al., 2002). Compared to the CTMI and the SQRT, a more accurate description of the $\sqrt{\mu_{\max}(T)}$ -relation is obtained with the aCTMI, certainly at temperatures below 30 °C. Also, the $T_{\min}$ estimate is more realistic, i.e., ≈ 6 °C, compared to 8–8.5 °C for the CTMI and SQRT. Use of the aCTMI improved $\sqrt{\mu_{\max}(T)}$ -data description which points at the existence of two phases in the suboptimal temperature region of *E. coli* K12. The alternation of the $\sqrt{\mu_{\max}(T)}$ is most likely related to the cold shock response. These results reveal a possible shortcoming of the model structure of commonly used secondary models describing the temperature effect on the microbial growth rate.

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1. Introduction

In general, each microorganism is characterized by its own intrinsic temperature region for growth, outlined by the minimum and maximum temperatures for growth ($T_{\min}$ and $T_{\max}$, respectively). At and below $T_{\min}$, growth is not possible and slightly above the minimum temperature for growth, the maximum specific growth rate, i.e., the (relative) increase in cell number as a function of time in the exponential phase ($1/N \, dN/dt$), is very low. As temperature increases, the rate of key intracellular processes increases and microorganisms grow faster. This relation only stands up to a certain temperature, i.e., T_{opt} , the temperature at which the maximum specific growth rate is maximal. Further increase of the temperature negatively affects the cellular metabolism and the growth rate decreases until $T_{\max}$ is reached. At and beyond this temperature, growth is no longer sustained. The maximum specific growth rate as a function of temperature is asymmetrically oriented around the optimum growth temperature T_{opt} . The increase in $\mu_{\max}$ when moving from $T_{\min}$ to T_{opt} is much slower than when moving from $T_{\max}$ toward T_{opt} . As such, the negative

effect of high temperatures is much more outspoken than the effect of low temperatures.

Within the domain of predictive microbiology, a series of secondary models exists to describe the influence of temperature on the microbial growth rate. According to Geeraerd et al. (2004), these secondary models can be subdivided into two groups. The first group consists of the square root models, the cardinal parameter model and the Arrhenius-based models, which include (some) biologically or graphically interpretable parameters. Square root models and cardinal parameter models are often extended toward more environmental factors via a multiplicative approach, i.e., the gamma concept. Furthermore, they are parsimonious and have a high fitting quality. The second group mainly consists of neural networks and response surface models. For the model structure selection, these models do not start from any a priori knowledge of the underlying relationship. A significant advantage of the lack of a fixed model structure is the high flexibility of these model types. All of the above models are empirical, data-driven black box models as no mechanistic information on the underlying mechanism of the temperature effect on the microbial metabolism is included.

The validity of the square root model (SQRT) developed by Ratkowsky et al. (1983) and the cardinal temperature model with inflection (CTMI) (Rosso et al., 1993) to describe the effect of temperature on the microbial growth rate is widely accepted. Together with the SQRT, the CTMI is among the most frequently used in microbiology (e.g., to describe biodegradation) and predictive microbiology. Up until now, exceptions have only been reported for *Listeria*, i.e., *Listeria*

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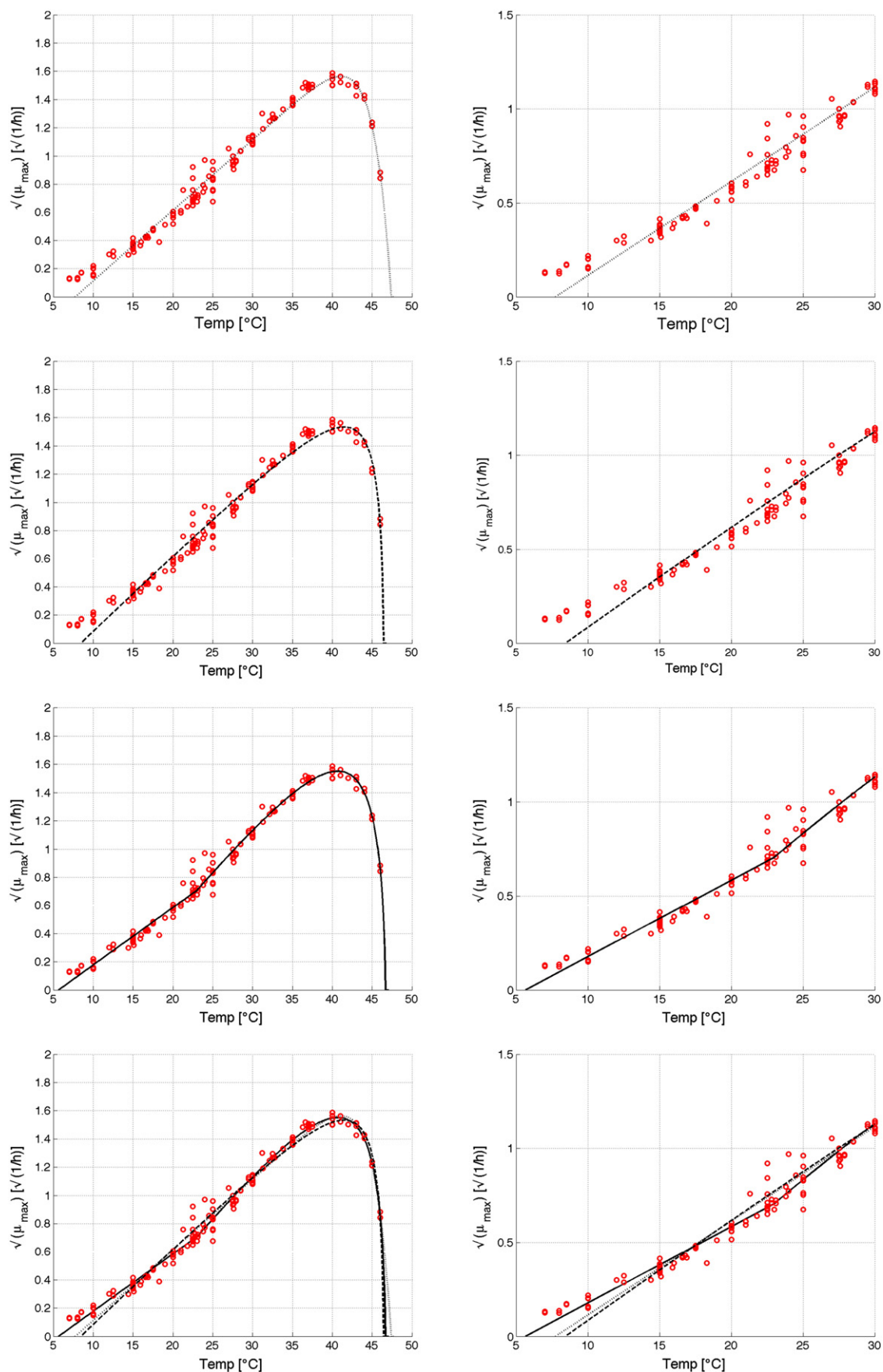


Fig. 1. Modeling the effect of temperature on the growth rate ($\sqrt{\mu_{\max}}(T)$) of *E. coli* K12 MG1655 with (row 1) the square root model (SQRT, Ratkowsky et al., 1983), (row 2) the cardinal temperature model with inflection (CTMI, Rosso et al., 1993), (row 3) the structural adaptation of the CTMI according to Le Marc et al. (2002) (aCTMI), and (row 4) a combination of all above simulations: (i) square root model ($\cdots$), (ii) model of Rosso ($---$), and (iii) model of Le Marc ($-$).

monocytogenes (Bajard et al., 1996) and *Listeria innocua* (Le Marc et al., 2002). In the suboptimal temperature region of these *Listeria* strains, the $\sqrt{\mu_{\max}}(T)$ -curve displays two linear phases. Inspired by the work of Bajard et al. (1996), Le Marc et al. (2002) added two parameters to the existing CTMI to model this deviating behavior, i.e., the change temperature T_c (as introduced by Bajard et al. 1996), and the temperature T_1 , which is the intersection between the first linear part of the model and the temperature axis.

The aim of this work is to evaluate the SQRT, CTMI and the adapted CTMI model for *Escherichia coli* K12 MG1655, with the focus on the description of its behavior at the suboptimal temperature region and near the minimum temperature for growth. An extended experimental data set, mainly collected from previous work studying the effect of temperature on the dynamics of *E. coli* K12 MG1655, was used to examine the relationship between temperature and the maximum specific growth rate for the selected microbial strain.

2. Material and methods

2.1. Bacterial strain

Experimental data shown were acquired with a culture of *E. coli* K12 MG1655 (CGSC#6300), obtained from the *E. coli* Genetic Stock Center at Yale University.

2.2. Experimental data

A total of 157 $\mu_{\max}(T)$ -data were extracted for the temperature region [7 °C 46 °C] from experiments collected from prior studies on the effect of temperature on the growth of *E. coli* K12 MG1655. This includes (i) test tube experiments (Van Derlinden et al., 2008a; Van Derlinden, 2009), (ii) bioreactor experiments (Bernaerts et al., 2002; Bernaerts, 2002; Swinnen, 2005; Swinnen et al., 2005, 2006; Van Derlinden et al., 2008b), (iii) Erlenmeyer experiments (Bernaerts, 2002; Van Derlinden, 2009), and (iv) unpublished data obtained using the protocols reported in the above references.

Maximum specific growth rate estimates were obtained by fitting experimental growth curves with the growth model of Baranyi and Roberts (1994). For experimental data collected under dynamic conditions, only data from static phases were considered. In this case, the growth rate was calculated as the slope of the corresponding part of the exponential phase.

In all cases, *E. coli* K12 MG1655 was grown in Brain Heart Infusion broth (BHI) and cell counts were determined from plating BHI-diluted samples on BHI agar. Possible differences in growth rates as a result of the different types of vessels, which can result in different oxygen concentrations, can be assumed minimal, as for all set-ups, aerobic conditions were guaranteed. It was observed from experiments performed at the same temperature using different setups that growth rates are highly similar.

The lower boundary of the temperature region considered was selected such that differentiation between growth and lag/no-growth is clear ($T \geq 7$ °C). For the upper boundary, temperatures higher than 46 °C are not included as (i) a sigmoidal growth dynamics were observed, or (ii) growth was not supported. At temperatures exceeding 44.5 °C, disturbed growth curves have been observed (Van Derlinden et al., 2008a). In this study, it was also shown that a pre-incubation at 45 °C for 24 h resulted in (approximately) sigmoidal growth curves at 45 and 46 °C. Growth rates obtained from these curves have been added to the data set in this paper.

2.3. Mathematical modeling

Three models, used to describe the influence of temperature on the microbial growth rate, are discussed.

(i) The square root model (Ratkowsky et al., 1983)

$$\sqrt{\mu_{\max}} = b \cdot (T - T_{\min}) \cdot (1 - \exp(c \cdot (T - T_{\max}))) \quad (1)$$

with b [$1/(\text{°C} \cdot \sqrt{\text{h}})$] and c [$1/\text{°C}$] constants, and $T_{\min}$ [°C] and $T_{\max}$ [°C] the notional minimum and maximum temperatures for growth, which can differ a few degrees from the observed $T_{\min}$ and $T_{\max}$.

(ii) The cardinal temperature model with inflection (Rosso et al., 1993)

$$\mu_{\max} = \gamma \cdot \mu_{\text{opt}} \quad (2)$$

with

$$T \leq T_{\min} \quad \gamma = 0$$

$$T_{\min} < T < T_{\max} \quad \gamma = \frac{(T - T_{\min})^2 (T - T_{\max})}{(T_{\text{opt}} - T_{\min}) \left((T_{\text{opt}} - T_{\min}) (T - T_{\text{opt}}) - (T_{\text{opt}} - T_{\max}) (T_{\min} + T_{\text{opt}} - 2T) \right)}$$

$$T \geq T_{\max} \quad \gamma = 0. \quad (3)$$

The parameters included in this model are the three cardinal temperatures $T_{\min}$ [°C], T_{opt} [°C], and $T_{\max}$ [°C] (i.e., the minimum, optimum and maximum temperatures for growth, respectively) and μ_{opt} [1/h] (i.e., the maximum specific growth rate at T_{opt}). Again, the values of $T_{\min}$ and $T_{\max}$ are notional values which can differ from observed cardinal values.

(iii) The structural adaptation of the CTMI by Le Marc et al. (2002). As a response to the observation of a nonlinear relation between $\sqrt{\mu_{\max}}$ and temperature for *Listeria* strains by Bajard et al. (1996), the original structure of the CTMI was adapted.

$$p(T) = \begin{cases} \frac{(T - T_1)^2 (T - T_{\max})}{(T_{\text{opt}} - T_1) \left((T_{\text{opt}} - T_1) (T - T_{\text{opt}}) - (T_{\text{opt}} - T_{\max}) (T_1 + T_{\text{opt}} - 2T) \right)} & \text{if } T \geq T_c \\ \frac{(T - T_{\min})^2 (T - T_{\max})}{(T_{\text{opt}} - T_1) \left((T_{\text{opt}} - T_1) (T - T_{\text{opt}}) - (T_{\text{opt}} - T_{\max}) (T_{\text{opt}} + T_{\max} - 2T_c) \right)} & \text{if } T \geq T_c \end{cases} \quad (4)$$

Next to the three cardinal temperatures and μ_{opt} , two additional parameters define the model, i.e., T_c and T_1 . T_c [°C] is the so-called change temperature and T_1 [°C] is the point of intersection between the first linear part and the temperature axis. By introducing the change temperature, the effect of lower temperatures on the growth rate, which was higher than predicted by the CTMI, could be modeled.

2.4. Data processing

Secondary model parameters and standard deviations were obtained by minimizing the sum of squared errors (SSE) using the *lsqnonlin* routine of the optimization toolbox of Matlab (The Mathworks, Inc.). The three models are fitted to the square transformed growth rate data, i.e., $\sqrt{\mu_{\max}}(T)$, to obtain a constant variance.

3. Results and discussion

3.1. Evaluation of the secondary models

A large set of $\mu_{\max}(T)$ -estimates was collected from previously performed static (or static parts from dynamic) experiments. Growth rate estimates were obtained by fitting (part of) the growth curves with the primary model of Baranyi and Roberts (1994). The collection of $\sqrt{\mu_{\max}}(T)$ -estimates was fitted both with the cardinal temperature model with inflection (CTMI) (Rosso et al., 1993) and the square root model (SQRT) developed by Ratkowsky et al. (1983). It has been reported that the fitting quality of the CTMI and the extended square root model of Ratkowsky et al. (1983) are comparable, but the CTMI

Within the normal physiological range, steady state levels of proteins are approximately constant. Adaptation of the metabolism is mainly obtained by changing the enzyme activity rather than the concentration (Herendeen et al., 1979). However, when the temperature is outside the NPTR, microorganisms encounter a temperature shock and protein levels change significantly. When *E. coli* is exposed to a low temperature, growth is stopped during a short phase in which the cells adapt to the new temperature. During this acclimatization phase, general protein synthesis is reduced and only a specific set of proteins, i.e., cold shock proteins (CSPs), is expressed (see, e.g., Giuliodori et al., 2004; Polissi et al., 2003). More specifically, a cold shock results in the decrease in and inhibition of the synthesis of proteins, which affect some key intracellular processes, e.g., RNA translation and transcription (Jones and Inouye, 1994; Phadtare et al., 1999; Thieringer et al., 1998). Multiple CSPs have been identified which intervene at different locations: energy generation metabolism, DNA supercoiling, stability of RNA and stability of proteins (Yamanaka, 1999). According to Ferrer et al. (2003), a specific type of stress induced proteins, i.e., chaperonins, determines the lower growth limit of *E. coli* K12. At low temperatures, the functionality of these chaperonins, i.e., protein (re)folding, decreases and growth is no longer sustained. As such, the minimum growth temperature of *E. coli* K12 seems to be related to the protein folding capability of the cells. Strocchi et al. (2006) identified that, at temperatures below the boundary of balanced growth, functionality of proteins is decreased as a result of the sub-optimal temperature. At these temperatures, general protein functionality is maintained via the interaction with chaperonins. When the ability of the chaperonins to stabilize proteins is insufficient, the minimum growth temperature is reached.

As the lower boundary estimate of the NPTR for *E. coli* K12 MG1655 (22.78 °C–23.86 °C; Swinnen et al., 2005) is very similar to the estimated change temperature T_c (23.0 °C), the presence of the cold shock proteins possibly results in a slower decline of the growth rate with decreasing temperature. Similarly, the lower boundary for *L. monocytogenes* has been estimated at 12 to 13 °C by Nichols et al. (2002). As for *E. coli*, this value corresponds with the change temperature T_c for *Listeria* strains, which has been situated between 10 and 15 °C by Bajard et al. (1996) and Le Marc et al. (2002).

The possible relation between the T_c and the NPTR is illustrated in Fig. 2, which shows the two linear phases observed in the Arrhenius plot for [7 °C 37 °C]. A sequence of two lines was fitted to the natural logarithm of the growth rate plotted as a function of $1/T$ (temperature expressed in Kelvin), with the intersection between the two straight lines fixed at T_c , i.e., 23 °C. This yields an accurate description, which seems to support the hypothesis that the T_c is related to a cold adaptation, i.e., a switch in the microbial metabolism and/or an adaptation of the membrane fluidity.

The above results, together with the earlier observation of Bajard et al. (1996), reveal a possible shortcoming of commonly used secondary models (SQRT and CTMI) that describe the microbial growth in the suboptimal temperature region. This work shows that the dynamics in the suboptimal temperature region, previously only observed for *Listeria* strains, also apply for *E. coli* K12 MG1655. As these dynamics are now already observed for two different strains, it becomes more likely that other strains reveal similar dynamics. For both strains, the change in the $\mu_{\max}(T)$ can be related to the normal physiological temperature range and the cellular response to low temperatures. As such, similar observations might exist for strains that know some kind of cold shock response. Further research, however, is required before this observation can be generalized. As other studies focusing on the effect of temperature on the maximum specific growth rate observed a relation that can be described accurately by the traditional CTMI and SQRT models (Rosso et al., 1993; Zwietering et al., 1994), the strain specificity of this a-typical behavior must be investigated. Also, new studies performed with other microbial strains for which specific cold responses have been

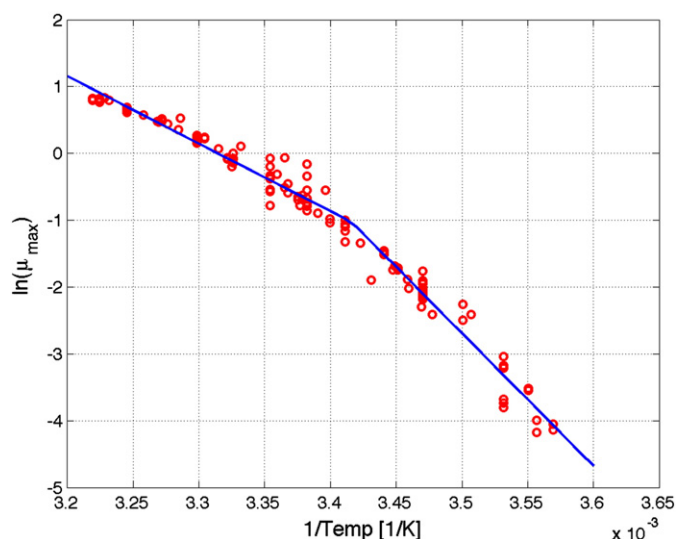


Fig. 2. Arrhenius plot of *E. coli* K12 in BHI broth with temperature expressed in Kelvin. Fit of a sequence of two straight lines with the inflection point equal to the T_c as given in Table 1 (—).

observed, e.g., *Salmonella*, will bring more light to this phenomenon. These studies should consider a high number of temperatures, specifically within the suboptimal temperature region.

Finally, it has to be mentioned that taking a linear relationship is a valid approach when only temperatures in the region $[T_{\min} T_c]$ are considered. When discussing the microbial dynamics during chilled storage, this temperature region is the most interesting. Only when higher temperatures are relevant, e.g., abuse chilling temperatures or storage at room temperature, the discussion about the correct model structure emerges.

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